

Combinatorial Algorithms (Math 482), 2026 Spring Problem Set 8

Due by Wednesday, March 25, 2026.

1. **Strongly connected components.** Let $G = (V, E)$ be a directed graph. We define a binary relation: For any two vertices $u, v \in V$, we say that $u \sim v$ if G contains a directed path from u to v , as well as a directed path from v to u . A *strongly connected component* is a maximal vertex set $W \subseteq V$ such that for any two vertices $u, v \in W$, it holds that $u \sim v$.

- (a) Find the strongly connected components of a directed cycle C_n and path P_n .
- (b) For every pair of integers $1 \leq k \leq n$, construct a directed graph $G_{n,k}$ that has precisely n vertices and k strongly connected components.

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2. **Graph Coloring.** We can color a graph $G = (V, E)$ by assigning a *color* $c(v)$ to each vertex v such that for every edge uv , we have $c(u) \neq c(v)$. The color $c(v)$ can be just a positive integer $1, 2, 3, \dots$ (so we never run out of colors). A *k-coloring* of a graph is a coloring with colors $\{1, \dots, k\}$.

- (a) Prove that a graph is 2-colorable if and only if it is bipartite.
- (b) Prove that if every vertex has at most Δ neighbors in a graph $G = (V, E)$, then G is $(\Delta + 1)$ -colorable.
- (c) For every integer $\Delta \geq 2$, construct a graph G_Δ such that the maximum degree of G_n is Δ but G_n is not Δ -colorable.

3. **Shortest path trees.** Let $G = (V, E)$ be a connected edge-weighted graph, where the weight of an edge is $w(e) > 0$. Fix a vertex $u \in V$. For every vertex $v \in V$, let $P(u, v)$ be a shortest path from u to v in G . Let $T(u)$ denote the union of the shortest paths $P(u, v)$ for all v . If $T(u)$ is a tree, it is called the **shortest path tree** of u .

- (a) Is it possible that there are two different shortest paths between the same pair of vertices, u and v , in a graph?
- (b) Construct an edge-weighted graph G such that $T(u)$ contains a cycle for some vertex u (and, in particular, $T(u)$ is not a tree).
- (c) Show that for every vertex u , one can choose shortest paths $P(u, v)$, for all $v \in V$, such that $T(u)$ is a tree.

4. **Shortest shortest-path tree.** Dylan suggests the following greedy algorithm for the *minimum spanning tree problem*: We are given a connected graph $G = (V, E)$ with positive edge weights. For every vertex u , compute a shortest-path tree $T(u)$, as described in Problem 3c above. Let $W(u)$ denote the total weight of $T(u)$. Find a vertex $u^* \in V$ that minimizes $W(u^*)$, and return $T(u^*)$.

Either prove that Dylan's algorithm finds an MST, or construct a connected edge-weighted graph where Dylan's algorithm returns a tree that is heavier than the MST.